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1. INTRODUCTION:

1.1 Climatic Impact & Waste Challenge:

With 241 million people, Pakistan is the fifth most populated country in the world. It produces about 38 million tons of solid garbage annually (Iqbal et al., 2023), and its yearly growth rate is more than 2.4% (Chatha et al., 2025). Large amounts of Municipal Solid Waste (MSW) have been produced as a result of rapid socioeconomic growth, necessitating the improvement of inadequate and inefficient waste management systems in order to reduce risks to the environment and public health.

Solid Waste Management (SWM) has thus turned into a socioeconomic crisis (Rashid et al., 2025). Primary and secondary collection are the two stages of municipal garbage collection in Pakistan (Mahmood & Khan, 2019). Roughly 90% of the waste that is collected ends up in open dumping sites, where it is neglected and no resources are recovered (Atta et al., 2020). Just 60% of waste in densely populated cities is collected, and uncollected MSW poses a major risk to public health through disease vectors, urban flooding, and environmental damage (Mousavi et al., 2023).

Since methane (CH_4) emissions from MSW are one of the main causes of greenhouse gas (GHG) emissions (Sohoo et al., 2022), ineffective treatment adds a significant amount of emissions to the environment. nearly 32 million tons (Mt) of CO_2-eq are produced annually by Pakistan's waste sector, including sludge waste. This amounts to nearly 6% of the country's total emissions, or 585 Mt of CO_2-eq (UNFCC-BUR-1, 2022). The Lahore Lakhodair disposal facility, which contributes 12.8% of GHG to the overall city level emissions, is a truly massive example. Unsettlingly, this trend is getting worse as CH_4 levels at Lakhodair have increased 15-fold since 2015. This is a dire situation that calls for policymakers to act quickly to find ways to turn this environmental problem into a climate funding opportunit (Maasakk et al., 2022). Pakistan's emissions can be reduced by sectoral laws and waste landfill diversion plans that focus on recycling recovery, composting, anaerobic digestion (AD), and final disposal with landfill gas (LFG) recovery (Devadoss et al., 2021).

1.2 Institutional Policy Challenges:

It is extremely difficult for policymakers to maintain the sector in Pakistan in the lack of any official national or provincial policies. The 18th Amendment gives all provincial governments the independent authority to use Local Government and Community Development (LG&CD) ministries to finance and carry out sanitation services. For FY 2025–2026, the province governments have set aside about PKR 221 billion for the trash industry. The government of Punjab, which has a 68% national population share, spends PKR 150 billion; the government of Sindh, which has a 20% population share, spends PKR 43 billion; the government of Khyber Pakhtunkhwa (KPK), which has an 8% population share, spends PKR 17.3 billion; and the government of Baluchistan, which has a 4% population share, spends PKR 8 billion. In the lack of any waste diversion legislation, strategies, or policies, the waste industry is still viewed as a liability while spending a reasonable sectoral budget, or 1.26% of the national total budget, because of the large investment and labor. Despite its efforts to reduce waste handling costs and emissions, the current outsourcing model in Punjab, Sindh, Baluchistan, and Islamabad completely ignores the role of this stakeholder. The informal waste sector is recovering an average of 14.4% of 10 recyclables from the municipality waste stream. The outsourcing models' lack of involvement with pertinent stakeholders is a reflection of the state-run trash sector's limited institutional capabilities.

1.3 Clean Green Pakistan Initiative:

Chief Minister Maryam Nawaz Sharif inaugurated the "Clean Green Pakistan" (Clean Punjab) program, the biggest province-wide sanitation effort in Pakistani history, with a rigorous 90-day deadline to turn Punjab into a totally waste-free area. This all-encompassing approach creates a consistent, integrated waste management system that gives equal attention to large cities and isolated rural communities, in contrast to earlier cleanup initiatives that primarily targeted major urban areas. The government has shifted from typical municipal operations by establishing public-private partnerships and working with specialist private sector firms to handle large-scale trash disposal in order to maximize efficiency and guarantee long-term sustainability.

Manual street sweeping, unclogging clogged sewage drains, removing hazardous open waste piles, and organizing door-to-door trash collection from private dwellings are just a few of the everyday localized operations that the project spearheads. This campaign's broad influence goes well beyond simple aesthetics, acting as a vital intervention for regional economies and public health. The program significantly reduces airborne pollutants and disease-carrying vectors like mosquitoes and rats by getting rid of rotting trash and standing water, which improves living conditions and raises the level of living for millions of people. Additionally, rehabilitating public parks, historical landmarks, and public gathering places fosters a welcome atmosphere that increases tourism in the area and strengthens civic pride.

By generating thousands of new job possibilities for administrative staff, supervisors, and sanitation workers throughout the province, this enormous operational expansion also serves as an economic engine. The implementation mainly depends on a contemporary, data-driven monitoring structure instead of human paperwork to provide transparency and keep the project moving forward. Using picture-based check-in and check-out systems, field workers and supervisors record their daily attendance using user-friendly digital dashboards and specialized mobile applications. Geotagged data and time-stamped photos are used to follow every cleanup effort in real-time, providing indisputable digital evidence of the conditions at trash sites before and after. High-level government representatives are able to keep a close eye on developments, hold local teams responsible, and make sure the province effectively meets its cleanliness goals thanks to this stringent technical control.

2. Background:

In many developing nations, environmental management plans frequently neglect MSW management (Batool et al., 2008). Waste management is a complex issue that necessitates a rigorous evaluation of its effects on the environment, technical factors, manner of execution, and financial implications depending on treatment and disposal methods. This significant shift has been brought about by urgent environmental issues, resource shortages, the fact of climate change, and a corresponding increase in knowledge and community involvement (Khisro et al., 2024). Making educated judgments on how to best reduce the likelihood of environmental dangers and maximize resource recovery requires baseline data on waste composition, waste generation, treatment, and disposal as well as climatic impacts.By delving deeply into the system's intricacies, we can suggest solutions that protect the environment while simultaneously making the most of our resources to realize its full potential (Majeed et al., 2018). SWM is a complicated field that calls for meticulous design, execution, and effective operation maintenance. Cost-effective strategies that align with environmentally responsible technologies should be taken into consideration (Iqbal et al., 2022b).

The threats to society posed by global warming and climate change are at an all-time high (Eckstein, 2021). Since the pre-industrial era, the concentrations of greenhouse gases (GHGs) in the atmosphere, such as carbon dioxide (CO_2), methane (CH_4), and nitrous oxide (N_2O), have reached dangerous levels. One of the main reasons for the concerning changes in the environment that we are witnessing is this sharp rise (Thomas et al., 2004). Two of the biggest issues facing contemporary society are climate change and global warming.One of the main causes of environmental issues is the rise in GHG emissions (D'Avignon et al., 2010). About 73% of the nation's CO_2-eq emissions from the waste sector come from the landfilling of MSW, with the remaining 27% coming from the treatment and disposal of industrial wastewater (Mir et al., 2017).

The "Clean Green Pakistan" effort has its roots in major regional disparities in waste management throughout Pakistan's most populous province, institutional fragmentation, and decades of severe, worsening sanitation challenges. In the past, only a few large cities, such as Lahore, had access to efficient solid waste management and automated cleaning services. Tens of millions of people lived in large rural villages, low-income neighborhoods, and smaller towns, all of which were totally ignored, resulting in widespread open trash disposal and burning. The absence of a cohesive system led to hazardous increases in waterborne and vector-borne illnesses, toxic open waste piles, severely clogged sewerage systems, and declining air quality. Additionally, prior localized cleaning initiatives were viewed as intermittent, event-based cleanups devoid of long-term planning, accountability, and transparency, which allowed municipal corporations to suffer from systemic mismanagement and "ghost" staff irregularities. Chief Minister Maryam Nawaz Sharif initiated the program in March 2024 to radically change the system after realizing that the current state of affairs was unsustainable for both economic growth and public health.

The government institutionalized the Clean Green Pakistan Authority because the project was founded on the fundamental tenet that every citizen, regardless of location, has the right to cleanliness. This project's background is firmly rooted in a contemporary shift toward climate-smart urban governance, equity for rural communities, and the development of a circular economy that turns waste into sustainable value. It does this by replacing manual municipal tracking with an AI-driven, GPS-monitored digital system and by moving away from isolated operations to standardized public-private partnerships.

3. LITERATURE REVIEW:

3.1 Sustainable Municipal Solid Waste Management:

Solid waste management includes gathering, moving, treating, and securely disposing of waste in landfills. The ecology is greatly impacted by these activities, which also frequently place a financial and social strain on the community and the agencies in charge of solid waste management. Due to differing opinions and perceptions of the effects of waste management programs, local authorities dealing with garbage face issues, demonstrations, and public opposition. The environmental, social, and economic effects that stakeholders in Johor Bahru, Malaysia, perceive are evaluated in this paper.

With the use of super decision software, the stakeholders' opinions are assessed using the Analytical Hierarchy Process (AHP), a multi-criteria decision making analytical tool. When it comes to environmental aspects and implications, stream ecology, flora and fauna, habitat loss, land use, and air quality are ranked higher. The social elements that are most important include population growth, public collaboration, and public knowledge of health and safety. Economic variables and repercussions are dominated by regulations, landfill capacity, operating and maintenance costs, and capital costs. Based on the stakeholders' goals, four alternative disposal plans—landfilling, recycling, incineration, and composting—were put out and ranked. Recycling and incineration were favored over composting and landfilling as disposal methods.

3.2 Sectorial Strategies & Comparative Analysis:

While impoverished countries frequently deal with inadequate garbage collection, which leads to considerable methane emissions, developed countries prioritize waste avoidance, recycling, and waste-to-energy technology over landfills. Adopting composting, material recovery, and public-private partnerships that are tailored to local conditions to enhance sustainability and economic results are effective solutions for developing regions.

Punjab has better collection (70–80%) than other regions, according to an assessment of Pakistan's solid waste management using WasteAware criteria, however resource recovery (10–30% recycling) and environmental control are still lacking nationwide. Punjab has the most controlled dumpsites, but overall performance is hampered by inadequate infrastructure, poor financial viability, and the exclusion of the unofficial recycling industry.

3.3 E- Waste Management Problems:

E-waste, also known as waste electrical and electronic equipment (WEEE), is any electrical or electronic appliance, device, subassembly, or component that is discarded because the user deems it to be outmoded or useless. The government and people of India are deeply concerned about the increasing amount of electronic and electrical trash that is produced there. The already overworked waste management system is being strained by both domestic production and imports. The problem is made more difficult by the absence of government regulations and their lax enforcement. The environment and human health may be harmed by incorrect treatment of e-waste, which contains numerous hazardous India has relatively little environmentally friendly e-waste management because of a number of issues, including a lack of technology, a lack of funding, a small number of industry participants, recyclers' ignorance, etc. This chapter uses the analytic hierarchy process (AHP) to identify and prioritize the many environmental sound management (ESM) concerns in India. Experts were tasked with identifying and prioritizing the variables or obstacles. After that, we ran the AHP and determined the consistency ratio in addition to the relative weights of the various criteria.

3.4 Decentralization & AHP:

The Bundelkhand region, which lies in central India, has a large supply of renewable energy resources, yet its rural energy needs are primarily met by non-renewable energy sources. Because non-renewable energy resources are finite and contribute to both environmental damage and health issues, it is imperative that renewable energy sources be used. Using renewable technology in rural energy planning contributes to sustainable development. The nation's rural energy needs may be largely satisfied by localized renewable energy technologies. The identification and ranking of possible renewable energy solutions for this region's rural areas satisfies the study's goal.

The Analytical Hierarchy Process (AHP) is used to prioritize possible renewable energy technologies for this area. One of the most popular multi-criteria decision-making (MCDM) tools in sustainable energy planning is AHP, which was first introduced by Thomas L. Saaty. AHP could assist decision-makers in determining the best course of action and setting priorities. This study highlights the use of AHP in prioritizing the potential renewable energy technologies for rural areas in this region and identifies the various merits and demerits of these technologies, including anaerobic digestion, solar photovoltaic, and biomass gasification.

The study's findings demonstrate the potential of solar photovoltaic to contribute significantly to sustainable development in this area. They will also provide policymakers and researchers with guidance on how to analyze rural energy planning most effectively and efficiently.

3.5 Economy & Waste Management:

The average cost of collecting rubbish in metropolitan areas across the nation is PKR 4,794 per ton (Iqbal et al., 2022b). Due to inadequate urban design, low public awareness, and ineffective SWM infrastructure, 14 discarded heaps may be seen along roads and streets in all major cities. 50% of municipal budgets in underdeveloped nations are set aside for SWM (Aleluia & Ferrão, 2016). The nation's economy can benefit from the sustainability of the SWM sector in terms of reaching self-sufficiency in fertilizer and power/energy, i.e., producing compost and biogas from organic waste. The Clean Green Pakistan program, regional economics, and the Analytical Hierarchy Process (AHP) come together to offer a data-driven framework that may turn waste management from an expensive problem into a circular economy that is sustainable. This premier province-wide project, which unifies solid waste collection in 36 districts and 141 tehsils, is described on the Clean Green Pakistan Portal. The program uses AI analytics and GPS tracking to update service delivery by processing more than 50,000 tons of waste every day and hiring a "silent army" of 150,000 sanitation workers.

4. PROBLEM STATEMENT:

Pakistan's waste management industry, especially in Punjab, presents a significant obstacle in moving from a conventional linear disposal model to a sustainable "waste-to-value" circular economy, even with the execution of the "Clean Green Pakistan" project to modernize municipal services. Local authorities lack the decision-making tools necessary to maximize resource recovery in the absence of a rigorous, data-driven framework such as the Analytical Hierarchy Process (AHP) to assess and balance competing economic parameters (capital costs, public-private partnerships, fee structures) against environmental metrics (landfill diversion, emission reductions, leachate control). In order to turn waste management from a costly public health risk into a profitable, sustainable, and environmentally friendly economic engine, it is still imperative to evaluate the precise economic and environmental effects of the Clean Green Pakistan effort.

4.1 RESEARCH QUESTIONS:

- How well can Punjab's municipal waste sector be transformed from a linear disposal model into a circular economy that is both economically and environmentally sustainable through the Clean Green Pakistan Waste-to-Value Initiative?
- How can the selection of waste-to-value technologies based on regional economic, technical, and environmental factors be statistically optimized using the Analytical Hierarchy Process (AHP)?
- How does the Clean Green Pakistan initiative profit from trash while protecting the environment?

4.2 RESEARCH OBJECTIVES:

- To evaluate how the Clean Green Pakistan Authority creates a clean, green, and climate-resilient environment throughout the province by turning regular municipal garbage into profitable goods and sustainable energy.
- To assess how economically well the Clean Green Pakistan Garbage Tax works to build a self-sufficient financial model that lessens the need for government subsidies.
- To help municipal planners select the optimal waste-to-value technology by using the Analytical Hierarchy Process (AHP) matrix to scientifically balance economic costs against environmental advantages.

5. SIGNIFICANCE OF RESEARCH:

This study is essential to updating our understanding of waste management in Pakistan. Researchers have long noted that Pakistan lacks sufficient, current, and trustworthy statistics regarding the amount of waste produced and its destination. This study builds a new and accurate database that future scholars can use by examining the actual outcomes of the Clean Green Pakistan Authority Bill 2025. Furthermore, the Analytical Hierarchy Process (AHP), a popular mathematical model used globally to make difficult decisions, has seldom ever been tried in Pakistan.This study directly applies AHP to Punjab's unique combination of peaceful rural communities and congested megacities. It closes a significant gap in the academic literature by testing how these sophisticated decision-making algorithms actually function when coping with local government financial constraints and a lack of technical manpower.

In terms of the environment, this study immediately aids Pakistan in combating climate change and shielding nearby residents from hazardous pollution. Approximately 20% of all methane gas produced by humans worldwide, which traps heat and warms the globe, comes from decomposing rubbish in open fields. In order to help Pakistan reach its international climate targets under the Paris Agreement, this study demonstrates how the Clean Green Pakistan project may significantly reduce these damaging greenhouse gases by processing rubbish rather than letting it decompose.

Lastly, by figuring out how to assist the thousands of unofficial garbage pickers that roam the streets every day, this study focuses on the human aspect of waste management. Currently, 10% to 20% of the province's plastic, paper, and glass are sorted out and recycled entirely by these unappreciated workers on their own, but they do so in hazardous, unsanitary conditions.

6. RESEARCH METHODOLOGY:

In order to determine the most economical and environmentally responsible approach of managing waste under the Clean Green Pakistan initiative, this research methodology integrates real-world data with

expert perspectives. In order to determine how much waste is produced and whether locals are prepared to pay the new monthly fees, which range from PKR 200 to PKR 5,000, the study first gathers data from two distinct locations: smaller rural communities and major metropolis like Lahore. We next distribute a specific survey to thirty to forty-five government financial planners, environmental scientists, and trash experts.

Based on local finances and environmental safety, these specialists assess and rank various waste solutions, such as heavy waste-to-energy facilities versus simpler organic composting hubs, using a clever mathematical system known as the Analytical Hierarchy Process (AHP). In order to verify mathematical correctness, determine the precise amount of money the province can save, and estimate the amount of dangerous greenhouse gas that can be stopped from escaping into the environment, all of this data is finally processed by computers.

7. RESEARCH DESIGN:

The research design is Mixed-Methods Research Design that integrates qualitative policy analysis with quantitative mathematical modeling in order to appropriately assess the economic and environmental effects of the Clean Green Pakistan initiative**.**

Data Collection Methods:

Structured questionnaires will be distributed to the service users to gather primary data.

Secondary data will be collected from the Government documents/ publications, Academic Journals, books etc.

Data Analysis:

Data collected will be statistically analyzed with the use of frequency distribution, percentage, Mean scores and descriptive analysis using software like SPSS.

Data Sources:

Primary Data

Primary data will be collected through questionnaires.

Secondary Data:

Secondary data will be obtained from:

Government report

Academic journals

Research articles

Official websites

Clean Green Pakistan "Green Clean Environment"

Location: ________________________

Property Type: [ ] Residential (House) [ ] Commercial (Shop/Business)

1.**How often is trash collected from your property?**

oDaily

o2–3 times a week

oRarely / Never

2.**How satisfied are you with the overall cleanliness of your neighborhood?**

oHighly Satisfied

o Satisfied
o Dissatisfied

3. **What is the biggest environmental problem caused by trash in your area?**

o Bad smells and disease
o Burning trash smoke
o Clogged drains and flooding

4. **Are you willing to separate your food waste from dry plastics/papers at home?**

o Yes
o No
o Only if free bins are provided

5. **What is the maximum monthly fee you are willing to pay for reliable daily waste collection?**

o PKR 200 – 500
o PKR 501 – 1,000
o PKR 1,001+
o Nothing

8. CONCLUSION:

The evaluation of the Clean Green Pakistan Waste-to-Value Initiative shows that it is both economically and environmentally feasible to create a Clean Green Environment by converting municipal solid waste management from a conventional dumping system into a structured circular economy [Urban Unit, 2025; Wilson et al., 2015]. The study provides quantitative evidence that moving away from open dumping directly reduces major urban risks, including air pollution from open burning, toxic leachate poisoning in groundwater, and catastrophic monsoon flash floods brought on by clogged drainage networks with plastic. By capturing high-organic municipal waste stream materials, the program addresses the regional anthropogenic methane (CH_4) emissions dilemma and contributes significantly to climate mitigation on a macro level [Clark & Paskin, 2022].

Economically, the study demonstrates that the industry can become financially self-sufficient over the long run and lessen its significant dependency on government assistance. A baseline citizen willingness to pay for the recently implemented Clean Green Pakistan user fee model (PKR 200 to 5,000+) is confirmed by analyzing the data from the public survey, as long as doorstep collection continues to be consistent and regular. Additionally, using the Analytical Hierarchy Process (AHP), decentralized approaches—like organic composting hubs and Material Recovery Facilities (MRFs)—produce the highest overall strategic ranking for secondary tehsils because of their high organic waste composition and low capital requirements, whereas high-tech Waste-to-Energy systems are ideal for major urban hubs supported by foreign investment [Li et al., 2015].

In the end, institutionalizing the informal garbage industry is essential to creating a truly sustainable clean green environment in Punjab [Nawaz et al., 2021]. In addition to significantly increasing resource recovery rates for the circular economy, formalizing local rubbish pickers into official, regulated green jobs also raises labor safety and public health standards. The Suthra Punjab approach provides emerging countries with a scalable blueprint for transforming municipal trash liabilities into self-sustaining environmental assets by effectively connecting mathematical decision-modeling (AHP) with localized policy reforms [Ajibade et al., 2019].

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